

JEE MAIN REVISION MASTER: COORDINATION COMPOUNDS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 319 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	IUPAC Nomenclature & Werner's Coordination Theory	217 PYQs	Coordination Basics
Very High	Crystal Field Theory (CFT) & Isomerism	189 PYQs	CFT & Isomerism

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
IUPAC Nomenclature & Werner's Coordination Theory	$\text{Primary valency (ionizable, satisfies charge)}, \quad \text{Secondary valency (non-ionizable, directional, satisfies CN)}$
Crystal Field Theory (CFT) & Isomerism	$\Delta_t = \frac{4}{9}\Delta_o, \quad \text{CFSE}_{\text{oct}} = [-0.4 n_{t_{2g}} + 0.6 n_{e_g}]\Delta_o + n_P, \quad \mu = \sqrt{n(n+2)} \text{ BM}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 IUPAC Nomenclature & Werner's Coordination Theory (Very High Yield — 217 PYQs)

Core Principle: Formulation of complexes, Werner's postulates, precipitable silver chloride fractions, and systematic IUPAC naming conventions.

■ **Caution / JEE Pitfall:** In $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$, adding excess AgNO_3 precipitates ONLY 2 moles of AgCl per mole of complex, because one Cl^- is coordinated.

■ **Speed Trick:** Chelating ligands (e.g. en, EDTA) form 5- or 6-membered rings with the central metal ion, imparting exceptional thermodynamic stability (chelate effect).

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Which of the following sequences of hybridisation, geometry and magnetic nature are correct for the given coordination compounds? A. $[\text{NiCl}_4]^{2-}$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Identify the correct statements from the following A. $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ is the most stable complex among $[\text{Fe}(\text{OH})_6]^{3-}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Match the LIST-I with LIST-II List-I Complex ion List-II Calculated spin only magnetic moment (BM) A. $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ I. 3...." [Ans: N/A]

3.2 Crystal Field Theory (CFT) & Isomerism (Very High Yield — 189 PYQs)

Core Principle: d-orbital splitting under octahedral and tetrahedral fields, high-spin vs low-spin complexes, spectrochemical series, and optical/geometrical isomerism.

■ **Caution / JEE Pitfall:** Tetrahedral complexes NEVER form low-spin complexes because $\Delta_t = (4/9)\Delta_o < P$ (splitting is always less than pairing energy).

■ **Speed Trick:** cis-[Co(en)₂Cl₂]⁺ is optically ACTIVE (non-superimposable mirror image), whereas the trans isomer has a plane of symmetry and is optically INACTIVE.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Which of the following sequences of hybridisation, geometry and magnetic nature are correct for the given coordination compounds? A. $[NiCl_4]^{2-}$ " **[Ans: N/A]**
- **[2026 Evening Shift] Q#1 (MCQ):** "Identify the correct statements from the following A. $[Fe(C_2O_4)_3]^{3-}$ is the most stable complex among $[Fe(OH)_6]^{3-}$, $[Fe(C_2O_4)_3]^{3-}$ " **[Ans: N/A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "Match the LIST-I with LIST-II List-I Complex ion List-II Calculated spin only magnetic moment (BM) A. $[Cr(H_2O)_6]^{2+}$ I. 3...." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

Given below are two statements : Statement I : Crystal Field Stabilization Energy (CFSE) of $[Cr(H_2O)_6]^{2+}$ is greater than that of $[Mn(H_2O)_6]^{2+}$. Statement II : Potassium ferricyanide has a greater spin-only magnetic moment than sodium ferrocyanide. In the light of the above statements, choose the correct answer from the options given below :

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are true **[Correct]**
- (C) Both Statement I and Statement II are false
- (D) Statement I is false but Statement II is true

Correct Answer: B

Solution Analysis: Statement I: In $[Cr(H_2O)_6]^{2+}$, Cr^{2+} is d^4 and with H_2O (weak field) it is high spin : $t_{2g}^3 e_g^1$. $CFSE = 3(-0.4\Delta_o) + 1(+0.6\Delta_o) = -0.6\Delta_o$ In $[Mn(H_2O)_6]^{2+}$, Mn^{2+} is d^5 high spin: $t_{2g}^3 e_g^2$. $CFSE = 3(-0.4\Delta_o) + 2(+0.6\Delta_o) = 0$ So CFSE of the chromium complex is greater (more stabilizing). Statement I is true. Statement II: $K_3[Fe(CN)_6]$: Fe^{3+} is d^5 ; CN^- is strong field $\rightarrow$ low spin t_{2g}^5 , 1 unpaired electron $\rightarrow \mu_{(so)} = \sqrt{1(1+2)} = \sqrt{3}$ BM. $Na_4[Fe(CN)_6]$: Fe^{2+} is d^6 ; CN^- is strong field $\rightarrow$ low spin t_{2g}^6 , 0 unpaired electron $\rightarrow \mu_{(so)} = 0$ BM. **[Ans: B]**

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Morning Shift (Q#2)

The correct statement among the following is:

- (A) $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$ are diamagnetic and $\text{Ni}(\text{CO})_4$ is paramagnetic.
- (B) $\text{Ni}(\text{CO})_4$ is diamagnetic and $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are paramagnetic.
- (C) $\text{Ni}(\text{CO})_4$ and $[\text{NiCl}_4]^{2-}$ are diamagnetic and $[\text{Ni}(\text{CN})_4]^{2-}$ is paramagnetic.
- (D) $\text{Ni}(\text{CO})_4$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $[\text{NiCl}_4]^{2-}$ is paramagnetic. **[Correct]**

Correct Answer: D

Solution Analysis: Option D is correct. 1) $\text{Ni}(\text{CO})_4$ Oxidation state of Ni: CO is a neutral ligand, so Ni is in 0 state.

Electronic configuration: $\text{Ni}(0) \Rightarrow 3d^{10}4s^0$ Since it is d^{10} , all electrons are paired. Hence, $\text{Ni}(\text{CO})_4$ is diamagnetic.

2) $[\text{Ni}(\text{CN})_4]^{2-}$ Oxidation state of Ni: $x + 4(-1) = -2 \Rightarrow x = +2$ So, Ni^{2+} is $3d^8$. CN^- is a strong field ligand, causes pairing of electrons. For $\text{Ni}^{2+}(d^8)$ with strong field ligands, complex becomes square planar (inner orbital), and all electrons get paired. Hence, $[\text{Ni}(\text{CN})_4]^{2-}$ is diamagnetic. 3) $[\text{NiCl}_4]^{2-}$